Properties of Fourier Transformation
Discrete Fourier Transform

$f(x)$ some function

$\Rightarrow \quad \mathcal{F}(f) \equiv \hat{f}$

Fourier

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \; e^{-i\omega \cdot x} \; f(x)$$

Inverse Fourier

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{i\omega \cdot x} \; \hat{f}(\omega)$$

examples:

$$\int e^{a \cdot x} = \frac{1}{a} e^{a \cdot x}$$

- $f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$

$$\Rightarrow \quad \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} dx \; e^{-i\omega \cdot x} = \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-i\omega} e^{-i\omega \cdot x} \right]_{-1}^{1} =$$

$$= \frac{i}{\sqrt{2\pi} \cdot \omega} \left(e^{-i\omega} - e^{i\omega} \right) = \frac{2 \cdot \sin \omega}{\sqrt{2\pi} \cdot \omega} = \sqrt{\frac{2}{\pi}} \; \frac{\sin(\omega)}{\omega}$$

- $f(x) = \delta(x)$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \; \delta(x) \, e^{-i\omega \cdot x} = \frac{1}{\sqrt{2\pi}}$$

inverse:

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{i\omega \cdot x} \underbrace{\frac{1}{\sqrt{2\pi}}}_{\hat{f}(\omega)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \; e^{i\omega \cdot x}$$

often used to represent $\delta(x)$

A few

Fourier transforms

$f(x)$	$\mathcal{F}(f)$
$1/\sqrt{2\pi}$	$\delta(\omega)$
$e^{-\lvert x\rvert}$	$\sqrt{\frac{2}{\pi}}\ \frac{1}{1+\omega^2}$
$\Theta(x)\, e^{-x}$	$\frac{1}{\sqrt{2\pi}}\ \frac{1}{1+i\omega}$
$e^{-\frac{x^2}{2}}$	$e^{-\omega^2/2}$
$\chi_{[-1,1]}(x)$	$\sqrt{\frac{2}{\pi}}\ \frac{\sin(\omega)}{\omega}$

Properties of F.T.

- linear

$$\mathcal{F}(a\cdot f + b\cdot g) = a\,\hat{f} + b\cdot\hat{g}$$

proof: denote $a\cdot f(x) + b\cdot g(x) \equiv F(x)$, $\mathcal{F}(F) \equiv \hat{F}$

$$\hat{F}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\; e^{-i\omega.x}\, F(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\; e^{-i\omega.x}\left(a\cdot f(x) + b\cdot g(x)\right)$$

$$= a\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\; e^{-i\omega.x}\, f(x) + b\cdot\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\; e^{-i\omega.x}\, g(x)$$

$$= a\cdot\hat{f}(\omega) + b\cdot\hat{g}(\omega) \qquad \square$$

- $g(x) \equiv f(-x) \quad\Rightarrow\quad \hat{g}(\omega) = \hat{f}(-\omega)$

proof:

$$\hat{g}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\; e^{\underbrace{-i\omega.x}_{(-\omega).(-x)}}\, f(\underbrace{-x}_{y}) =$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dy\; e^{-i(-\omega)\cdot y}\, f(y) = \hat{f}(-\omega) \qquad \square$$

Simple application:

$$g(x) = e^{-|x|} = \underset{\uparrow}{f(x)} + f(-x)$$

$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & x < 0 \end{cases} \Rightarrow$$

$$\hat{g}(\omega) = \hat{f}(\omega) + \hat{f}(-\omega) = \frac{1}{\sqrt{2\pi}}\left(\frac{1}{1+i\omega} + \frac{1}{1-i\omega}\right) = \sqrt{\frac{2}{\pi}}\ \frac{1}{1+\omega^2}$$

$$\mathcal{F}\left\{e^{-|x|}\right\} = \sqrt{\frac{2}{\pi}}\ \frac{1}{1+\omega^2}$$

exponential $\leftrightarrow$ Lorentzian

- general case ($a \in \mathbb{R}^+$)

$$\boxed{g(x) = f(a \cdot x) \Rightarrow \hat{g}(\omega) = \frac{1}{a} \cdot \hat{f}\left(\frac{\omega}{a}\right)}$$

<u>proof</u>:

$\omega \cdot \frac{1}{a} \cdot a \cdot x$

$$\hat{g}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\ e^{-i\omega x} \cdot f(\underbrace{a \cdot x}_{= \tilde{x}}) =$$

$d\tilde{x} = a \cdot dx$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d\tilde{x}}{a}\ e^{-i\frac{\omega}{a}\cdot\tilde{x}}\ f(\tilde{x}) \underset{\tilde{x} \to x}{=} \frac{1}{a} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\ e^{-i\frac{\omega}{a}\cdot x}\ f(x)}_{\hat{f}\left(\frac{\omega}{a}\right)}$$

$\square$

- if $f(x) \in \mathbb{R} \Rightarrow \hat{f}(\omega) = \hat{f}(-\omega)^*$

<u>proof:</u>

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{i\omega\cdot x} \hat{f}(\omega)$$

take complex conjugate $\Rightarrow$

$$f(x) = f^*(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{-i\cdot\omega\cdot x} \hat{f}(\omega)^* =$$

$\tilde{\omega} = -\omega$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\tilde{\omega} \; e^{i\tilde{\omega}\cdot x} \hat{f}(-\tilde{\omega})^* = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{i\omega\cdot x} \hat{f}(-\omega)^*$$

$\tilde{\omega} \to \omega$

subtract as

$$0 = f(x) - f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{i\omega\cdot x} \underbrace{\left(\hat{f}(\omega) - \hat{f}(-\omega)^*\right)}_{\emptyset} \qquad \square$$

for any x

- $\mathcal{F}(f') = i\omega \hat{f}(\omega)$

$f(x), f'(x)$ continuous $\quad f(x) \xrightarrow[|x|]{\infty} 0$, $|f'|$ integrable

<u>Proof:</u>

$$\mathcal{F}(f')(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \; f'(x) e^{-i\omega x} =$$

$$= \underbrace{\left[\frac{1}{\sqrt{2\pi}} f(x) e^{-i\omega\cdot x}\right]_{-\infty}^{\infty}}_{= \emptyset} - \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi}} f(x) \underbrace{\left(-i\omega \cdot e^{-i\omega\cdot x}\right)}_{\left(e^{-i\cdot\omega\cdot x}\right)'}$$

$$= i\omega \hat{f}(\omega) \qquad \square$$

- $$\mathcal{F}[f(x-b)] = \hat{f}(\omega)\cdot e^{-i\omega\cdot b}$$

P.:

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}dx\; \underbrace{e^{-i\omega x}}_{y+b}\; f(\underbrace{x-b}_{y}) = \left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}dy\; e^{-i\omega\cdot y} f(y)\right)\cdot e^{-i\omega\cdot b}$$

$\square$

convolution

$$h \equiv f * g$$

$$h(x) = \int_{-\infty}^{\infty}dy\; f(x-y)\,g(y) = \int_{-\infty}^{\infty}dy\; f(y)\,g(x-y)$$

sum $= x$!

- $f*g = g*f$
- $(f*g)*h = f*(g*h) = \ldots \equiv f*g*h$

Convolution theorem

$$\mathcal{F}[f*g] = \sqrt{2\pi}\cdot\hat{f}(\omega)\cdot\hat{g}(\omega)$$

→ important in probability theory

proof:

$$\sqrt{2\pi}\;\mathcal{F}[f*g](\omega) = \int_{-\infty}^{\infty}dx\; \underbrace{e^{-i\omega\cdot x}}_{e^{-i\omega(z+y)}}\int_{-\infty}^{\infty}dy\; f(\underbrace{x-y}_{z})\cdot g(y)$$

$$= \int_{-\infty}^{\infty}dz\int_{-\infty}^{\infty}dy\; \underbrace{e^{-i\omega\cdot z}\cdot e^{-i\omega\cdot y}\cdot f(z)\cdot g(y)}_{(e^{-i\omega z}f(z))\cdot(e^{-i\omega\cdot y}g(y))} =$$

$$= \left(\int_{-\infty}^{\infty}dz\; e^{-i\omega\cdot z}f(z)\right)\cdot\left(\int_{-\infty}^{\infty}dy\; e^{-i\omega\cdot y}g(y)\right) = 2\pi\;\hat{f}(\omega)\cdot\hat{g}(\omega)$$

$\square$

these rules can be combined ...

$$\mathcal{F}\left\{ e^{-(x-1)^2} \right\} = ?$$

$$\mathcal{F}\left\{ e^{-\frac{x^2}{2}} \right\} \Rightarrow e^{-\omega^2/2}$$

$$e^{-x^2} = e^{-\frac{(\sqrt{2}\cdot x)^2}{2}} \xrightarrow{F.T.} \frac{1}{\sqrt{2}} e^{-\frac{(\omega/\sqrt{2})^2}{2}} = \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}}$$

$$e^{-(x-1)^2} \xrightarrow{F.T.} e^{-i\omega} \cdot \frac{1}{\sqrt{2}} e^{-\frac{1}{4}\omega^2}$$